UNIT 10 · STANDARD MPP.4

Math is Ingenuity

Lesson 10-4

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes
Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use ratios and models to explain how the Penny Farthing and gear-driven bicycles were ingenious solutions to a real problem.

Language Objective: I can explain a gear ratio using the words rotation, teeth, and 'for each ____, the wheel turns ____.'

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

ingenuity | rotation | gear | ratio | equivalent ratios

👆 *Tap any word to see its meaning and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. Coming up with a different, sometimes unique solution is ______________.

2. One full turn of a wheel, gear, or pedal is a ______________.

3. A toothed wheel that passes motion through a chain is a ______________.

4. A comparison of two quantities, like pedal turns to wheel turns, is a ______________.

5. Ratios that make the same comparison with different numbers are ______________ ______________.

Write About the Math

How will your future vehicle be powered, and what math will be used in its design or use?

¿Con qué funcionará tu vehículo del futuro, y qué matemáticas se usarán en su diseño o su uso?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **ingenuity** *ingenio*
- ☐ **rotation** *rotación*
- ☐ **gear** *engranaje*
- ☐ **ratio** *razón*
- ☐ **equivalent ratios** *razones equivalentes*

Model: *One pedal rotation makes one wheel rotation, so a bigger wheel's longer perimeter carries the bicycle farther with each pedal push.* — A strong answer connects the 1-to-1 ratio to the wheel's perimeter, explaining a bigger wheel travels farther per pedal push. A weak answer just says 'bigger is faster' without naming the ratio or perimeter.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Describe a future mode of individual transportation. A complete answer says how it is ____, names a ____ (or rate) that shapes it, and explains where that math appears in the vehicle's ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.

- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.